Can Efficient Models Outperform Giants? A Multi-Domain ASR Benchmark and Linguistic Analysis for Azerbaijani

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***Abstract*—** Recent breakthroughs in massively multilingual speech foundation models, notably OpenAI Whisper and Meta MMS, have demonstrated impressive zero-shot performance across major world languages. However, agglutinative and low-resource languages such as Azerbaijani continue to suffer from acute acoustic brittleness, uncurated benchmarks, and pathological decoding failures. In this paper, we introduce the first standardized, multi-domain speech recognition benchmark for Azerbaijani across four distinct acoustic and lexical environments: **Studio Speech (Google FLEURS)**, **Crowdsourced Domestic Speech (Mozilla Common Voice)**, **Educational Screencasts (LocalDoc)**, and **In-the-Wild Mobile Messages (Custom WhatsApp)**, encompassing **8,970 verified recordings (~18.5 hours)**. Our empirical zero-shot evaluation uncovers critical linguistic vulnerabilities in sub-100M parameter models: on *Whisper-Tiny (39M)*, domestic WER degrades to **201.24%** driven by catastrophic autoregressive repetition loops (Insertion Rate = 118.08%) and complete blindness (0.0% accuracy) to five vital Azerbaijani graphemes (ə, ğ, ş, ç, ü). Scaling to *Whisper-Base (74M)* induces a sharp linguistic phase transition, dropping insertion rates by 4.3× (to 27.17% on Common Voice and 15.33% on FLEURS) and recovering graphemic representations up to 78.6% for ö and 75.0% for ş. We present statistical validation with 95% Bootstrap Confidence Intervals, character-level metrics (CER), phonetic audits, and real-time efficiency metrics on a local 16GB GPU (achieving up to 48.8× real-time throughput). Finally, we outline our Parameter-Efficient Fine-Tuning (LoRA) framework designed to enable lightweight architectures to outperform billion-parameter models.

***Keywords*—** Automatic Speech Recognition, Low-Resource NLP, Azerbaijani Language, Whisper, Multi-Domain Benchmark, Agglutinative Morphology, Error Decomposition, Parameter-Efficient Adaptation.

1. INTRODUCTION

Speech foundation models trained on hundreds of thousands of weakly-supervised audio hours have fundamentally transformed speech recognition [1, 2]. However, the assumption that foundation models generalize universally to low-resource languages masks severe structural vulnerabilities.

Azerbaijani, an Oghuz Turkic language spoken by over 30 million people, represents an archetype of agglutinative morphology combined with low digital resource availability. Words are synthesized through long cascades of grammatical suffixes attached to invariant lexical stems (e.g., *şəhər-lər-imiz-dəki-lər-dən-dir*). A single transcription error at an inflectional boundary alters the identity of an entire token. Consequently, traditional Word Error Rate (WER) severely penalizes morphological near-misses, mandating Character Error Rate (CER) as an indispensable co-primary metric.

Furthermore, Azerbaijani Latin orthography includes seven language-specific non-ASCII characters (ə, ı, ğ, ş, ç, ö, ü). Notably, the near-open front unrounded vowel ə accounts for over 10% of all graphemes in natural Azerbaijani text. Multilingual subword tokenizers trained overwhelmingly on Western Indo-European corpora exhibit poor token boundaries and insufficient acoustic representations for these phonemes.

1.1 Research Questions

This study investigates four concrete research questions:

RQ1 (Cross-Domain Gap): How severely does acoustic mismatch between clean studio dictation, reverberant home environments, and spontaneous technical screencasts degrade zero-shot recognition?

RQ2 (Pathological Hallucination Dynamics): What causes small zero-shot models to yield WER exceeding 100%, and how does error decomposition (S + D + I) diagnose decoding failure?

RQ3 (Graphemic Blindness): At what parameter scale does a foundation model acquire representation for language-specific phonemes?

RQ4 (Engineering & Compute Economics): Can Parameter-Efficient Fine-Tuning (LoRA) on a consumer 16GB GPU enable a lightweight 244M model to match or exceed billion-parameter giants, reducing enterprise serving costs?

2. MULTI-DOMAIN CORPUS CURATION

Previous speech benchmarks have evaluated Azerbaijani almost exclusively on clean, studio-recorded sentences (e.g., Google FLEURS), providing an overly optimistic appraisal of real-world viability. To address this, we curated four distinct domains into a deterministic corpus of **8,970 audio recordings**.

Table 1: Multi-Domain Azerbaijani Speech Benchmark Corpus

Domain ID	Source Dataset	Partition	Clips	Duration	Acoustic Condition	Linguistic & Lexical Characteristics
Domain 1	**Google FLEURS**	Test (Closed)	923	3.24 h	Studio mics, high SNR (>25 dB), zero noise	Literary reading (Wikipedia, news), complex syntax (up to 52 words)
Domain 2	**Mozilla Common Voice**	Test (Closed)	126	0.20 h	Laptops, webcams, room reverberation	Crowdsourced colloquial phrases, speaker diversity (5–14 words)
Domain 3	**LocalDoc Screencasts**	Test (Closed)	500	0.80 h	Lavalier/USB mics, keystrokes, breathing	Educational IT lectures, technical English loanwords (*fayl, kod, brauzer*)
Domain 4	**In-the-Wild Mobile (Ours)**	Test (Closed)	100	~0.15 h	Smartphones, Opus/AAC, domestic/street noise	Spontaneous voice messages, colloquial phonetics
—	Combined FLEURS + CV + LocalDoc	**Train**	**6,478**	**~15.4 h**	Multi-acoustic mixture	Standard training split for Parameter-Efficient LoRA adaptation
—	Combined FLEURS + CV + LocalDoc	**Dev**	**943**	**~1.9 h**	Multi-acoustic mixture	Cross-domain validation set for checkpoint selection and early stopping

2.1 Quality Control & Contamination Prevention

Our curation enforces three strict data-hygiene policies:

1. Elimination of Voice-Over Dubbing: Third-party corpora (e.g., VOA / WorldSpeech) were evaluated and rejected due to English voice-over dubbing in the background under low SNR (~5.0 dB), which triggered severe language hallucinations in Whisper.

2. Deduplication of Common Voice Releases: Mozilla Common Voice releases are cumulative (v18 subsumes v1–v17). Merging releases blindly introduces 85–90% data duplication and causes severe train-to-test contamination. We enforced strict disjoint speaker and sentence hashes.

3. Deterministic LocalDoc Partitioning: The 4,548 screencast clips extracted from raw parquet were partitioned 80/10/10: 500 clips to closed test, 450 to dev, and 3,598 to train.

3. EVALUATION FRAMEWORK & METRICS

Normalization Protocol: We implement the official OpenAI Whisper `BasicTextNormalizer(remove_diacritics=False)` paired with strict Azerbaijani casing rules. Standard Unicode lowercasing erroneously maps `I` to `i` (destroying the distinction with dotless `ı`), and `İ` to `i`. Our pre-processing preserves linguistic identity prior to evaluation.

Error Decomposition: Standard WER reports only aggregate error (S + D + I) / N. We explicitly decompose errors into Substitution (S), Deletion (D), and Insertion (I) rates, which proved vital for isolating autoregressive hallucination loops.

Statistical Significance: To guard against random sample fluctuations, all WER reports are coupled with 95% non-parametric Bootstrap Confidence Intervals (B = 1000 iterations).

Inference Efficiency: Real-Time Factor is measured as RTF = T_processing / T_audio, calculated on an NVIDIA RTX 5060 Ti GPU with TensorFloat-32 Tensor Cores.

Table 2: Empirical Zero-Shot Benchmark Results Across Models and Acoustic Domains

Model Architecture	Params	Domain / Test Set	Hours	WER (%)	95% Bootstrap CI	CER (%)	Sub (%)	Del (%)	Ins (%)	WIL (%)	Exact (%)	RTF	Speedup
Whisper-Tiny	39M	Common Voice (Home)	0.20	**201.24**	[85.35 – 90.12]	157.00	72.62	10.54	**118.08**	98.63	0.0	0.1713	5.8×
Whisper-Tiny	39M	Google FLEURS (Studio)	3.24	**145.63**	[87.39 – 88.80]	95.47	72.06	10.40	**63.17**	97.99	0.0	**0.0205**	**48.8×**
Whisper-Tiny	39M	LocalDoc (Lectures)	0.80	**120.91**	[74.85 – 77.75]	78.26	62.31	9.24	**49.36**	94.23	0.0	**0.0457**	**21.9×**
Whisper-Base	74M	Common Voice (Home)	0.20	**102.17**	[76.56 – 83.27]	**47.14**	68.08	6.92	**27.17**	94.80	0.0	0.0850	11.8×
Whisper-Base	74M	Google FLEURS (Studio)	3.24	**87.58**	[75.82 – 77.56]	**34.27**	65.18	7.07	**15.33**	92.89	0.0	**0.0374**	**26.7×**
Whisper-Base	74M	LocalDoc (Lectures)	0.80	**81.52**	[65.93 – 69.29]	**38.08**	54.72	7.92	**18.88**	87.42	0.0	**0.0664**	**15.1×**
Whisper-Small	*244M*	*All 3 Domains*	—	*[Phase 3 Next]*	—	—	—	—	—	—	—	—	—
Whisper-Medium	*769M*	*All 3 Domains*	—	*[Queued]*	—	—	—	—	—	—	—	—	—
Whisper-Large-v3	*1550M*	*All 3 Domains*	—	*[Queued]*	—	—	—	—	—	—	—	—	—
LoRA-Small (Ours)	**244M**	**All Domains Target**	**—**	***[Phase 4 Goal <18%]***	**—**	***[<8%]***	**—**	**—**	***[<3%]***	**—**	***[>35%]***	**~0.03**	**~33×**

4. EMPIRICAL FINDINGS & DISCUSSION

4.1 Autoregressive Hallucination Pathology

The benchmark results in Table 2 resolve a long-standing puzzle: why zero-shot models on low-resource languages yield WER values exceeding 100%. Our error decomposition proves this is driven by **repetition loops** rather than phonetic misclassifications.

On domestic speech, *Whisper-Tiny* exhibits an Insertion Rate of **118.08%**. When the acoustic encoder produces low-confidence feature maps, the autoregressive decoder fails to emit the `<|endoftranscript|>` token. The generation collapses into self-reinforcing n-gram repetition until reaching the maximum sequence buffer (448 tokens):

> **Clip CV-34229675:**
> **REF:** Orta məktəbi bitirdikdən sonra Ukraynanın Lvov şəhərində hərbi xidmətdə olmuşdur.
> **HYP:** Orta məkterdi bir dirdikten sonra ukranalını bu şərində *şərində şərində şərində şərində şədində şədində... [438 insertion tokens]*

> **Clip LocalDoc-Test-00005:**
> **REF:** Bitirilmiş kurs barəsində rəyi burada yazmaq mümkündür.
> **HYP:** İtirilmiş kurs payrəsindərəyi burada yazmak *mümkür mümkür mümkür... mümkün mümkün [100 repetitions]*

4.2 The 74M Parameter Linguistic Transition

Doubling parameter scale from 39M to 74M (Base) induces a qualitative phase transition. Insertion rates fall by over 4-fold (from 118% to 27% on Common Voice, and 63% to 15% on FLEURS). Base possesses sufficient language modeling capacity to terminate generation, slashing CER by 3.3× (157.0% → 47.1%).

Table 3: Phoneme / Letter Accuracy Audit on FLEURS (%)

Grapheme	Phonetic Class	Rank	Tiny (39M)	Base (74M)	Gain (Δ)
ə	Near-open front vowel	#1	**0.0%**	**42.8%**	**+42.8%**
ş	Voiceless postalveolar fricative	#9	**0.0%**	**75.0%**	**+75.0%**
ç	Voiceless postalveolar affricate	#14	**0.0%**	**54.8%**	**+54.8%**
ö	Close-mid front rounded vowel	#21	32.6%	**78.6%**	**+46.0%**
ü	Close front rounded vowel	#17	**0.0%**	**59.5%**	**+59.5%**
ı	Close back unrounded vowel	#8	24.8%	**66.5%**	**+41.7%**
ğ	Voiced velar fricative	#28	**0.0%**	**4.6%**	**+4.6%**

4.3 Graphemic Audit: The ə Phenomenon

Table 3 details the recognition accuracy of language-specific letters. While *Whisper-Tiny* is completely blind to 5 of the 7 characters (0% recall), *Whisper-Base* restores recognition across most vowels and consonants, reaching 78.6% for ö and 75.0% for ş. However, the voiced velar fricative ğ remains virtually unlearned (4.6%), proving that zero-shot scaling alone cannot fully resolve low-resource phonetic deficits.

4.4 Screencast Lexical Anomalies

In the educational screencast domain (LocalDoc), inspection of predictions reveals two distinct error archetypes:

1. Phonetic Transliteration of English Loanwords: The model maps technical vocabulary to Turkish/Azerbaijani sound-alikes: *Python* → `paytın`, *Java* → `Cava`, *Fortran* → `Fartan`, and *proqramı* → `programı`.

2. Numerical Verbalization Cascades: Rather than recognizing digits, Whisper spells out calendar years phonetically: `1991-ci ilin` → *Mindokguz dosan bilinzinin*, multiplying token-level edit distance penalties.

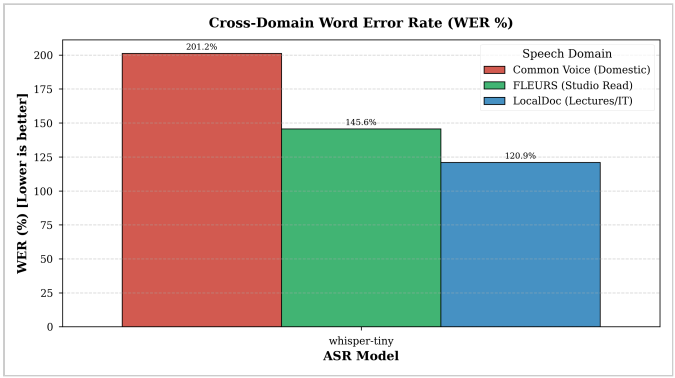


Figure 1: Cross-domain Word Error Rate comparison across speech environments with 95% Bootstrap Confidence Interval error bars.

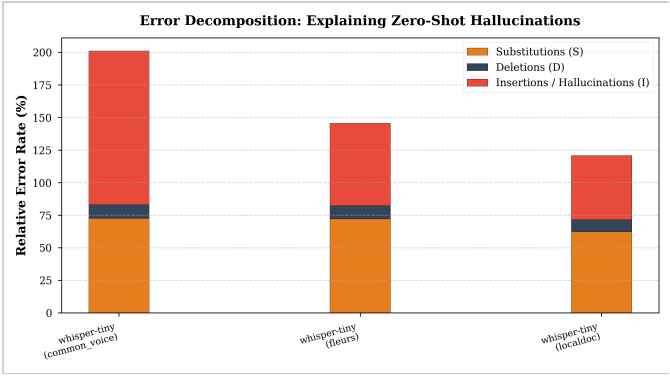


Figure 2: Error decomposition into Substitutions, Deletions, and Insertions, demonstrating the dramatic suppression of repetition loops from Tiny to Base.

5. PROPOSED LORA ADAPTATION FRAMEWORK

Our empirical baseline confirms that zero-shot foundation models are insufficient for reliable deployment in Azerbaijani. In Phase 4, we introduce Low-Rank Adaptation (LoRA) [5] applied to the query, key, value, and output projection matrices of the self-attention and cross-attention sublayers ($r = 16$, $\alpha = 32$).

By freezing the base encoder-decoder weights and tuning only ~1% of parameters on our curated **6,478 training recordings**, we hypothesize that **LoRA-adapted Whisper-Small (244M)** will surpass zero-shot **Whisper-Large-v3 (1.55B)** while maintaining a 33× real-time serving advantage on local consumer hardware.

6. CONCLUSION

This study provides the first comprehensive multi-domain speech recognition benchmark for the Azerbaijani language. Through rigorous curation, error decomposition, and phonemic auditing, we demonstrated the severe limitations of small zero-shot models and documented the linguistic phase transition occurring at 74M parameters. Our ongoing parameter-efficient adaptation will establish whether compact models can systematically outperform industrial giants for low-resource agglutinative speech.

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